

1. Administrative & Study Identification

Full Study Title: A Phase 1/2 Study of BMS-986340 as Monotherapy and in Combination With Nivolumab or Docetaxel in Participants With Advanced Solid Tumors **Short Title/Acronym:** A Study of BMS-986340 as Monotherapy and in Combination With Nivolumab or Docetaxel in Participants With Advanced Solid Tumors **Protocol Number:** CA052-002 **NCT Number:** NCT04895709 **Trial Status:** RECRUITING **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** BMS-986340 (anti-CCR8 monoclonal antibody), Nivolumab (anti-PD-1), Docetaxel (ATC Code: L01CD02; Trade names: Docefrez, Taxotere) **Phase of Development:** PHASE 1 / PHASE 2 **Medical Monitor:** Bristol-Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, tolerability, and recommended dose(s) of BMS-986340 as monotherapy and in combination with nivolumab or docetaxel in participants with advanced solid tumors. This study is a first-in-human (FIH) study of BMS-986340. **Secondary Objectives:** To determine the pharmacokinetic (PK) profile of BMS-986340 (including C_{max}, T_{max}, AUC) and to assess preliminary anti-tumor activity in the study population.

2.2 Background & Rationale To address the gap in treatment for patients with advanced or metastatic solid tumors who are refractory to standard-of-care therapies. While first-generation immune checkpoint inhibitors (like anti-PD-1) have transformed cancer care, resistance often occurs. BMS-986340 is an investigational immunotherapy targeting the CCR8 receptor. This First-In-Human (FIH) study investigates if administering BMS-986340 alone or in combination with nivolumab (immunotherapy) or docetaxel (chemotherapy) can safely enhance the immune system's ability to recognize and destroy tumor cells.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose Escalation & Expansion) **Blinding:** Open-label (None) **Randomization:** Non-Randomized (Sequential Assignment)

3.2 Planned Enrollment & Duration Target **NS:** 949 participants **Number of Sites:** Multicenter / Global **Estimated Study Start:** 27-May-2021 **Estimated Primary Completion:** Up to 120 weeks from study initiation per participant

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years. 2. Advanced or metastatic solid tumors with radiographically documented progressive disease on or after the most recent therapy. 3. Measurable disease per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1 and at least 1 lesion accessible for biopsy. Fine needle biopsy, cytology, and bone lesion biopsies are not acceptable. 4. Fresh pre-treatment and on-treatment tumor biopsy must be provided for biomarker analysis. 5. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 6. Must have received standard-of-care therapies, including an available programmed death (ligand)-1 inhibitor known to be effective in the tumor type for which they are being evaluated. (*Exception: Part 1C excludes participants with prior docetaxel use for the advanced/metastatic setting*). 7. Advanced or metastatic disease and have received, be refractory to, not be a candidate for, or be intolerant of existing therapies known to provide clinical benefit for the condition of the participant.

4.2 Exclusion Criteria 1. Women who are pregnant or breastfeeding. 2. Primary central nervous system (CNS) malignancy, untreated CNS metastases, or leptomeningeal metastases. 3. Active, known, or suspected autoimmune disease. 4. Condition requiring systemic treatment with either corticosteroids within 14 days or other immunosuppressive medications within 30 days of the first dose of study treatment. 5. History of or active interstitial lung disease or pulmonary fibrosis. 6. Concurrent malignancy requiring treatment or history of prior malignancy active within 2 years prior to the first dose of study treatment. 7. Uncontrolled or significant cardiovascular disease. 8. Major surgery within 4 weeks of study drug administration. 9. Prior organ or tissue allograft.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: BMS-986340 administered in sequential dose escalation (Parts 1A, 1B, 1C) followed by dose expansion cohorts (Parts 2A, 2B), either as monotherapy or combined with standard regimens of nivolumab or docetaxel. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Up to 120 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for advanced oncology patients. **Prohibited:** Systemic immunosuppressive medications and corticosteroids (unless for physiological replacement or managing adverse events), and concurrent investigational anti-cancer agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Tumor Biopsy (Pre-treatment), Baseline Imaging (RECIST v1.1), ECOG Assessment
Baseline	Day 0	First Dose (IV Infusion), Vital Signs, Safety Labs
Interim	Treatment Cycles	On-treatment Tumor Biopsy, Safety/DLT Assessments, PK blood sampling, Imaging every 8-12 weeks
End of Study	Up to Week 120	Final Imaging, End of Treatment Evaluation, AE/SAE Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability evaluated by the incidence of adverse events (AEs), serious adverse events (SAEs), and incidence of AEs meeting protocol defined dose-limiting toxicity (DLT) criteria. **Measurement Method:** Clinical laboratory assessments, physical examinations, and continuous protocol-defined toxicity monitoring.

7.2 Secondary & Exploratory Endpoints

- 1. Pharmacokinetics (PK):** Pharmacokinetic parameters of BMS-986340 administered as monotherapy: Maximum concentration (C_{max}), Time to maximum concentration (T_{max}), and Area under the concentration-time curve 1 dosing interval (AUC (TAU)).
- 2. Preliminary Efficacy:** Objective Response Rate (ORR), Progression-Free Survival (PFS), and Overall Survival (OS) based on RECIST v1.1.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of treatment-emergent adverse events (TEAEs) and dose-limiting toxicities (DLTs) from informed consent through 120 weeks.

Serious Adverse Events (SAEs): Required to be reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Safety Review Committee meetings are held to guide transition from dose-escalation to dose-expansion phases.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all AE and DLT assessments (any participant receiving ≥ 1 dose of study drug); Efficacy Evaluable Set for all participants with measurable disease and at least one post-baseline scan. **Sample Size Justification:** Estimated N = 949, structured using standard dose-escalation statistical models (e.g., 3+3 or Bayesian optimal interval) to accurately determine the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D). **Handling of Missing Data:** PK/PD

parameters and continuous variables will be analyzed using standard oncology trial imputation techniques.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard clinical site monitoring to ensure protocol compliance and safety reporting. **Audit Trail:** Secure Electronic Data Capture (eDC), strict adherence tracking for fresh biopsies, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: CA052-002 **Registry Number:** NCT04895709 **Posting ID:** 1058595903106701529 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study of BMS-986340 as Monotherapy and in Combination With Nivolumab or Docetaxel in Participants With Advanced Solid Tumors **Official Regulatory Title:** A Phase 1/2 Study of BMS-986340 as Monotherapy and in Combination With Nivolumab or Docetaxel in Participants With Advanced Solid Tumors

12. Clinical Framework (Oncology Specific)

Primary Conditions / Disease Areas: Cervical Cancer, Gastric/ Gastroesophageal Junction Adenocarcinoma, Microsatellite Stable Colorectal Cancer, Non-Small-Cell Lung Cancer, Squamous Cell Carcinoma of Head and Neck, Carcinoma, Renal Cell, Urothelial Carcinoma, Pancreatic Adenocarcinoma, Melanoma, Ovarian Neoplasms, Triple Negative Breast Neoplasms **Preferred UMLS Name:** Advanced Malignant Neoplasm **Diagnosis Threshold:** Radiographically documented progressive disease & measurable disease per RECIST v1.1 **Medical Methodology:** Targeted Immunotherapy (Anti-CCR8) + Anti-PD-1 (Nivolumab) or Chemotherapy (Docetaxel) **Optimization Target:** Maximum Tolerated Dose (MTD) / Recommended Phase 2 Dose (RP2D)

13. Study Procedures & Methodology

Management Pathway: Dose Escalation mapping (Parts 1A, 1B, 1C) transitioning to Dose Expansion (Parts 2A, 2B) **Education Protocol (HCP):** Site initiation visits covering study drug administration, DLT criteria, and fresh biopsy acquisition handling **Education Protocol (Patient):** Informed consent outlining the risks of first-in-human investigational drugs, multiple biopsies, and immunotherapy toxicities **Quality Audit Mechanism:** Central pathology review for tumor biopsies and rigorous independent radiologic review for RECIST endpoints

14. Observation & Follow-Up Schedule

Total Duration: Up to 120 weeks **Onsite Visit Cadence:** Required for all IV infusions, PK sampling, and fresh biopsies **Remote Monitoring Frequency:** Continuous pharmacovigilance follow-up **Checklist Review Interval:** Prior to every new treatment cycle

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					